ANA 322 Eyeball

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Eyeball: Has three coats

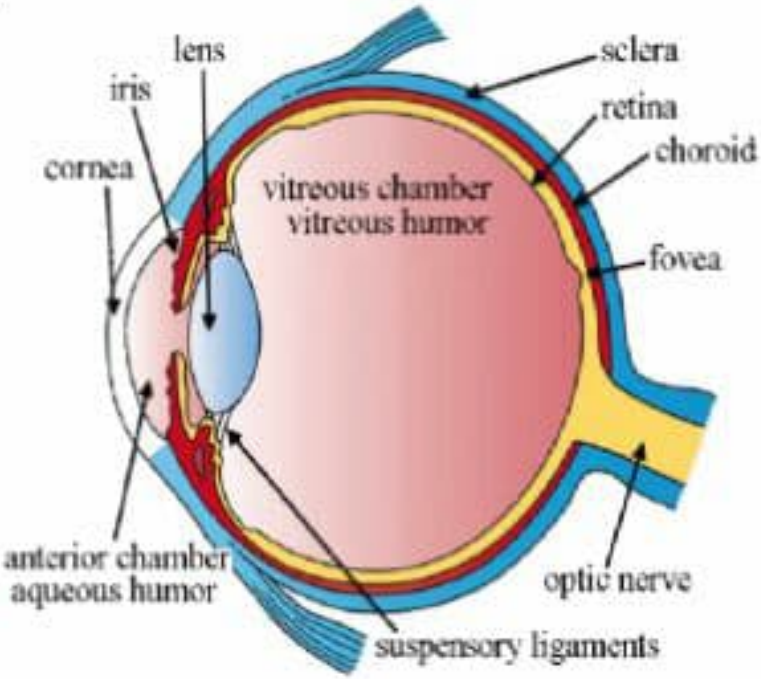
Outer fibrous- sclera and cornea.

Middle vascular- choroid, ciliary body and iris.

Innermost – retina.

Sclera: Opaque forms post. 5/6th of the ball. Non distensible tough fibrous coat. Thickest at entrance of optic nerve. Thinnest 6mm behind sclera-corneal junction.

- Outer of sclera white and smooth with Tenon capsule. The anterior by conjunctiva seen as white.
- Inner part brown grooved for ciliary nerve and artery and separated from choroid by perichoroidal space.







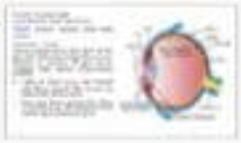



- Sclera ends anteriorly at sclera-cornea junction (Limbus). Deep part, circular canal of Schuemm at periphery anterior of chamber front of sclera-corneal junction.
- Aqueous humour through the canal to sclera vein.
- Sclera usually pierced by optic nerve, ciliary nerve and artery.
- Applied anatomy: Rupture of the eyeball usually at sclera-corneal junction.

Cornea: Transparent replacing sclera anteriorly from sclera-corneal junction and forming anterior 1/6th.

- Avascular, nourished by lymph circulating through numerous cornea spaces.
- Nerve supply; mainly short ciliary nerves and sensitive to only pain.

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Middle coat (uveal trait) - **Choroid**: Pigmented layer separating the posterior of sclera from retina.

- Brownish in man and blackish in many animals. Anteriorly terminates at Ora serrata merging with ciliary body.
- Between choroid and sclera- perichoroidal space rich in ciliary vessels and nerves.
- Attachment of sclera to choroid loose while to retina tight.
- It has 4 layers: (i) Suprachoroidal lamina- Contains collagen, elastic fibres and pigmented cells. (ii) Vascular lamina - Arteries, veins, pigment cells and short ciliary veins. (iii) Chorio- capillary lamina – Important, nourishes rods and cones cells by diffusion. (iv) Inner basal lamina (memb. Of Buch) – Thin transparent membrane firmly attached to outer pigment of retina.

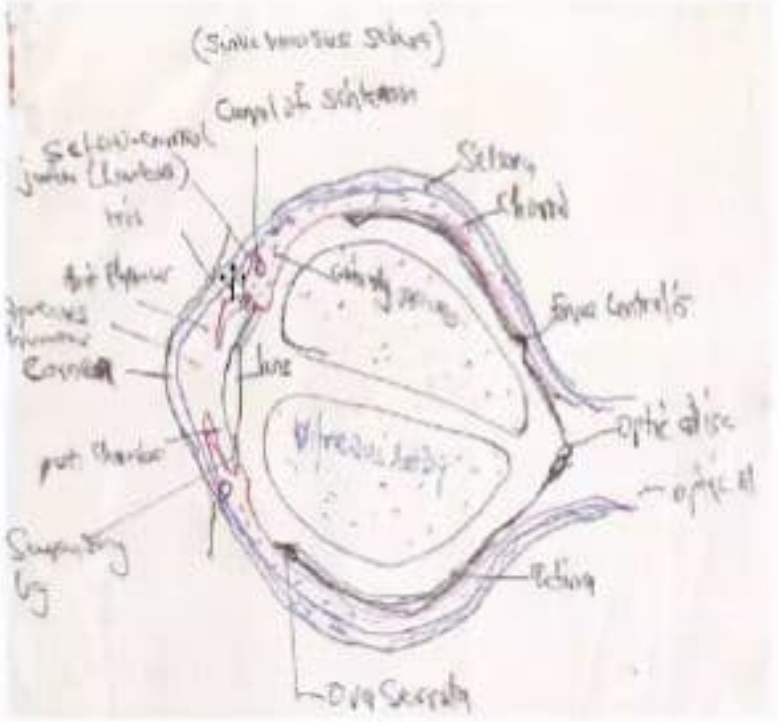
Ciliary body – Thickest part of uveal trait posterior to sclera-corneal junction. Posteriorly continues with choroid and anteriorly continues with iris.

- Gives attachment to lens and helps accommodate near vision.
- Applied to inner surface of sclera; thick in front and thin behind.
- Has ciliary muscle and processes.

Ciliary muscle – Smooth unstriated, helps focuses lens for near vision.

- Forms intrinsic muscle of eyeball, supplied by P∑ element of C_3.

When contracts suspensory lig. relaxes and lens bulges.



Iris: Constitute anterior of uveal trait. Forms circular curtain with central space – pupil. Adjusting size of pupil controls light entering the eye.

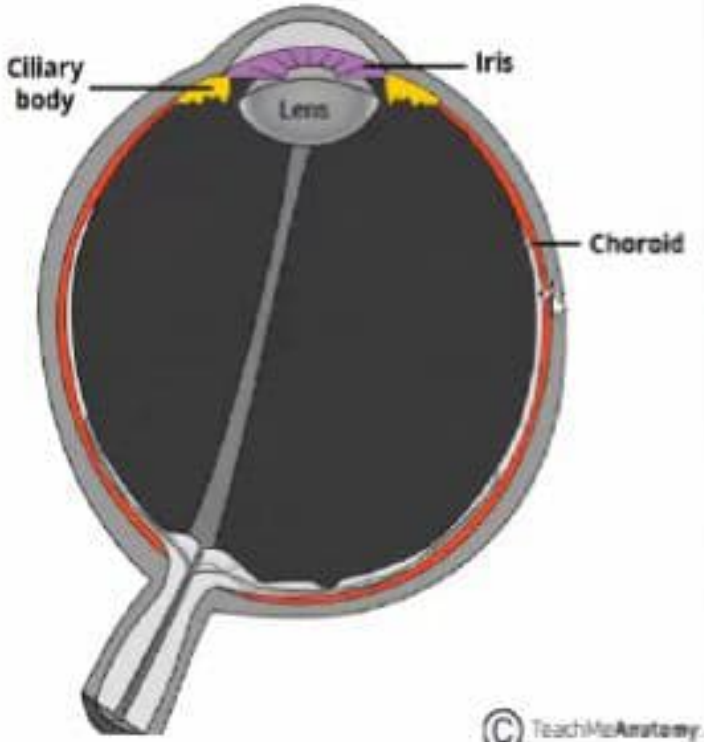
- Pupil size controlled by well developed circular muscle –sphincter pupillae, innervation PΣ and Ill defined radial muscle at posterior of iris, innervation Σ.

Iris divides anterior segment of eye into anterior and posterior chambers containing aqueous humour.

Retina- Innermost coat, neural layer of eyeball. Neural layer sensitive to light ends anteriorly at Ora serrata.

At entrance of optic nerve is circular area – optic disc 1.5mm diameter no rod and cone therefore not sensitive to light (Blind spot).

- 3mm lateral to optic disc is another depression macula lutea avascular and yellow in colour. Centre of another depression- Fovea centralis- thinnest part of eye sight of maximum visual acuity.

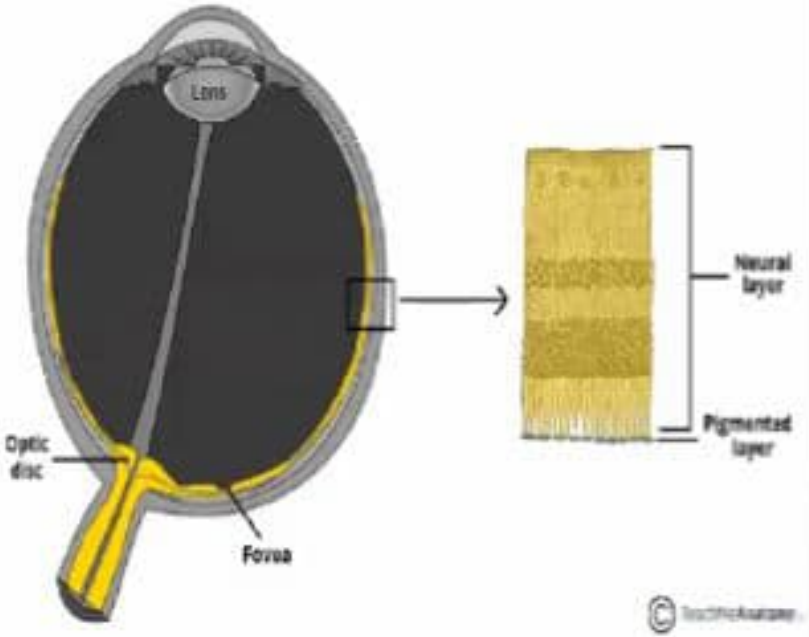


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Rods and cones – Cones contain idopsin sensitive to bright light and colours known as photopic vision.

Rod – contain visual purple and responds to dim light and black and white colours (scotopic vision).

Retina comprises ten layers: (i) outer pigmented layer, (ii) Layer of rods and cones, (iii) Ext limiting membrane, (iv) Outer nuclear layer, (v) Outer plexiform layer (vi) Inner nuclear layer (Bipolar cells), (vii) Inner plexiform layer, (viii) Ganglion cell layer, (ix) Nerve fibre layer, (x) inner limiting membrane.



- Three neuron interposes between retina and cortex –
- First neuron- bipolar cell of retina.
- Second neuron, - ganglionic cell of retina. Axon pass to lateral geniculate body.
- Third starts from lateral geniculate body to sensory cortex.

- Clinical considerations:
- Injury to cornea may cause opacities (cataract) that may interfere with vision. Congenital cataract caused by viral infection –rubella.
- Over production of aqueous humour or lack of its drainage may raise intra ocular pressure. A condition called glaucoma.
- Retina artery an end artery, blockage leads to sudden blindnes

There are two main clinical classifications of glaucoma:

Open angle – where the outflow of aqueous humor through the trabecular meshwork is reduced. It causes a gradual reduction of the peripheral vision, until the end stages of the disease.

Closed angle – where the iris is forced against the trabecular meshwork, preventing any drainage of aqueous humor. It is an ophthalmic emergency, which can rapidly lead to blindness.